

Introduction to machine learning with Watsonx

Advanced Practical Course

Working with LLMs

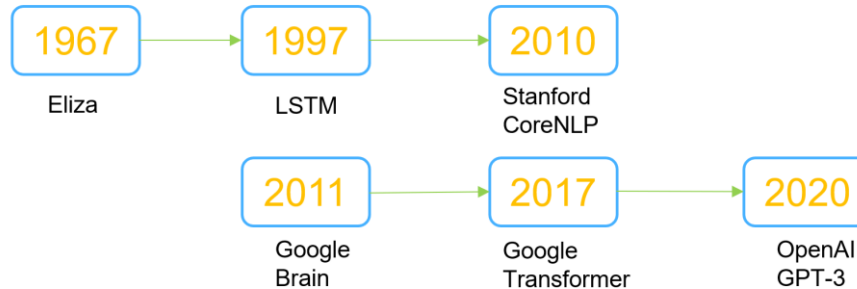
What is a Large Language Model?

- **Definition:** LLMs are a specialized form of generative artificial intelligence
- **Definition:** Generative AI refers to a class of artificial intelligence models that create new content by learning from existing data (for example text, image, video, voice).
- **How it Works:** It uses neural networks, especially deep learning models, to generate data that mimics the input data it's trained on.

Short History of NLP/LLM's

2022: Release of ChatGPT by OpenAI

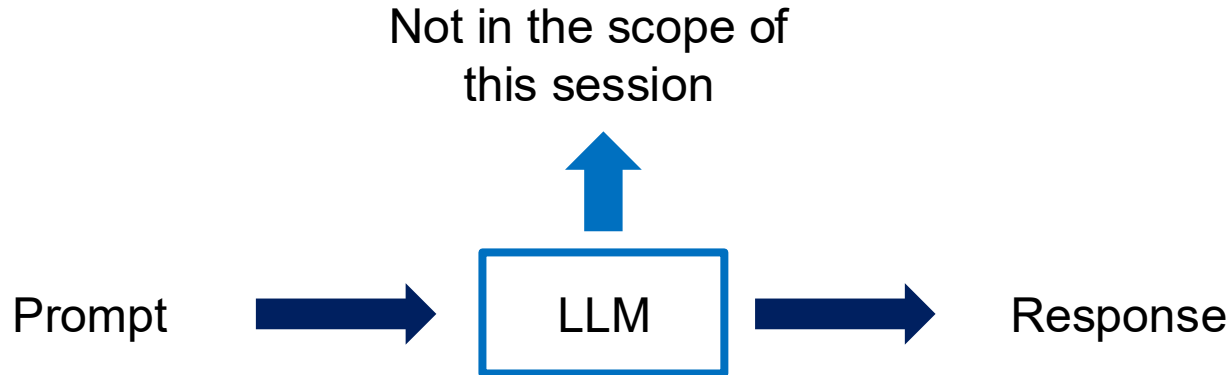
- Initial Capabilities: Text generation
- Today: image generation, “reasoning and problem-solving”.



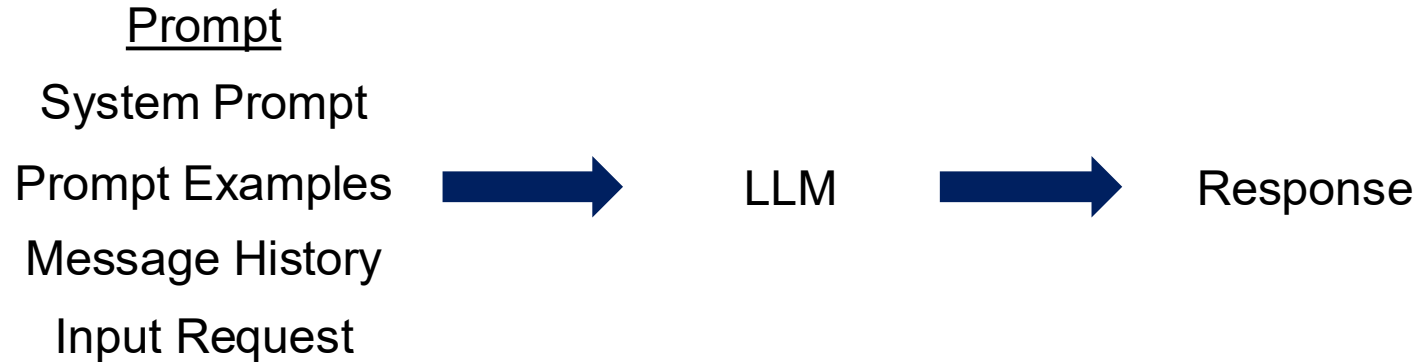
Overview Selected LLM Providers

- **GPT (OpenAI)**
- **Claude (Anthropic)**
- **Gemini (Google)**
- **Llama (Meta) – OpenSource**
- **Mistral (Mistral) – OpenSource(*)**
- **DeepSeek (DeepSeek) - OpenSource**

How to Interact with an LLM?



How to Interact with an LLM?



How to Interact with an LLM?

Prompt

System Prompt
Prompt Examples
Message History
Input Request

Prompt Strategy

Zero shot learning
One shot learning
Few-shot learning

Chain of Thought (CoT)

Zero shot learning

Prompt:

Classify the sentiment of the following text as positive, negative, or neutral.

Text: I think the vacation was okay. Sentiment:

Response:

?

Zero shot learning

Prompt:

Classify the sentiment of the following text as positive, negative, or neutral.

Text: I think the vacation was okay. Sentiment:

Response:

Neutral

One shot learning

Prompt:

Classify the sentiment of the following text as positive, negative, or neutral.

Text: The product is terrible. Sentiment: Negative

Text: I think the vacation was okay. Sentiment:

Response:

?

Few shot learning

System Prompt:

Classify the sentiment of the following text as positive, negative, or neutral.

Prompt Examples:

Text: The product is terrible. Sentiment: Negative

Text: Super helpful, worth it. Sentiment: Positive

Input Request:

Text: It doesnt work! Sentiment:

Response:

?

Chain of Thought

With Chain of Thought (CoT) Prompting the model should reason about the question first and then give the answer.



This could increase the probability that the model arrive at the correct answer

Chain of Thought

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

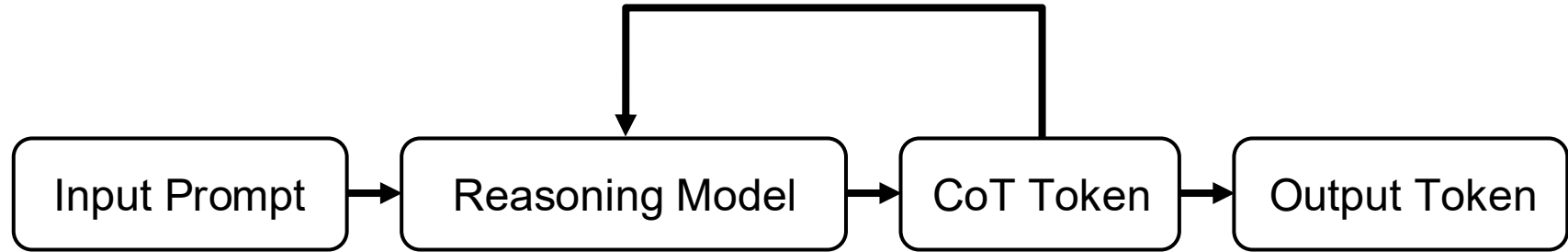
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

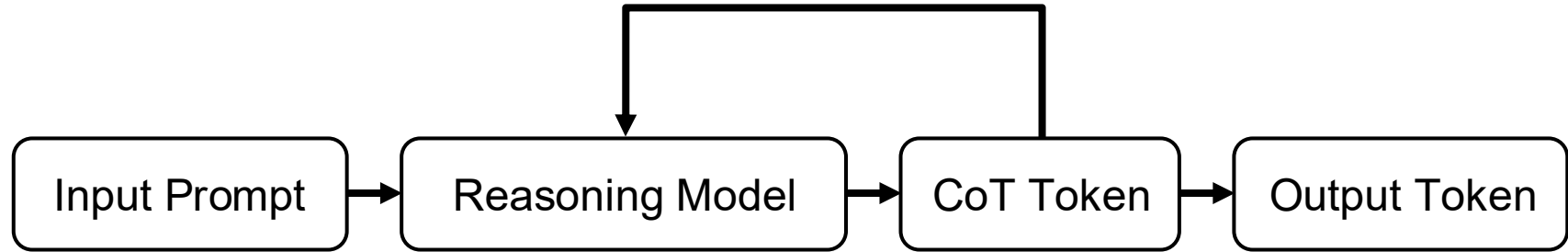
Chain of Thought – Reasoning Model



Example -> DeepSeek R1

Demo

Chain of Thought – Reasoning Model



Example -> DeepSeek R1

Demo

Training Compute Scaling

Models:

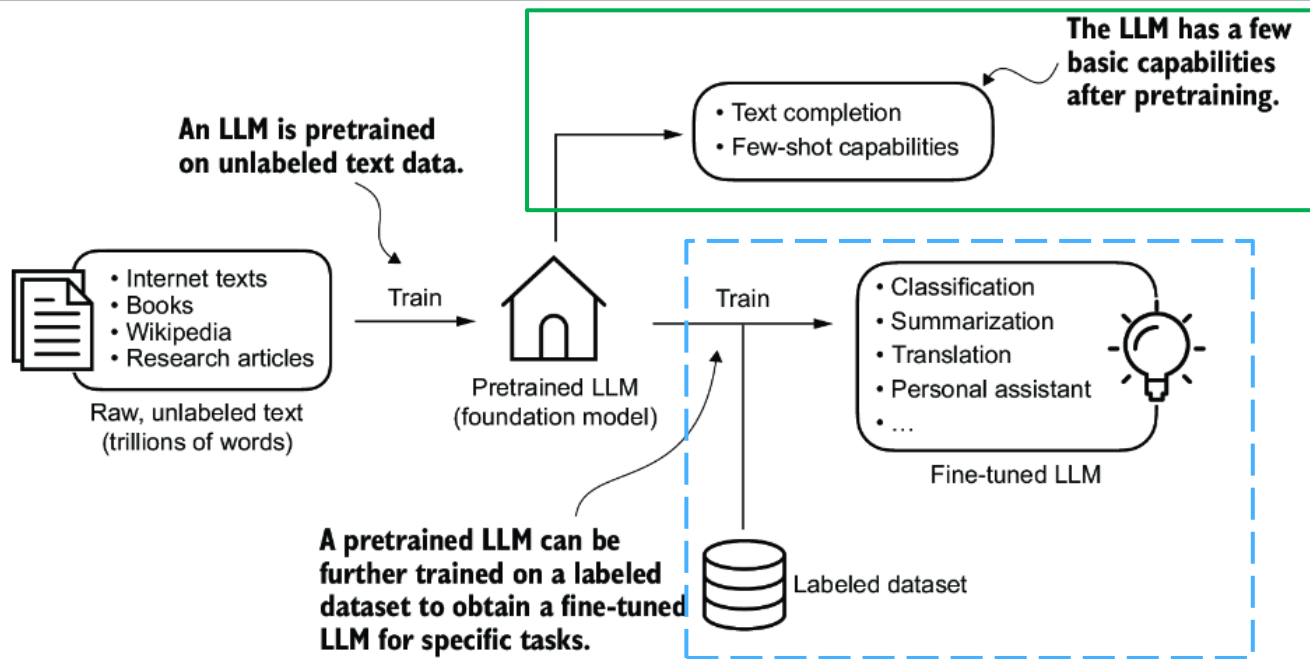
- GPT 4
- Lama
- Mistral

Test-Time Compute Scaling

Models:

- GPT-5
- R1
- Claude 4.7 Opus

LLM Functionality



S. Raschka, *Build a Large Language Model (From Scratch)*. Manning, 2024.

Bring our own data

How can we add custom
(e.g. company specific)
data to an LLM?



1. Context Extension
2. Retrieval Augment Generation
3. Finetuning
- (4. Tool Calling)

Context extension

„Easiest“ way is to add information directly to the prompt -> Extend the context



Context extension

„Easiest“ way is to add information directly to the prompt -> Extend the context



Demo + Exercise

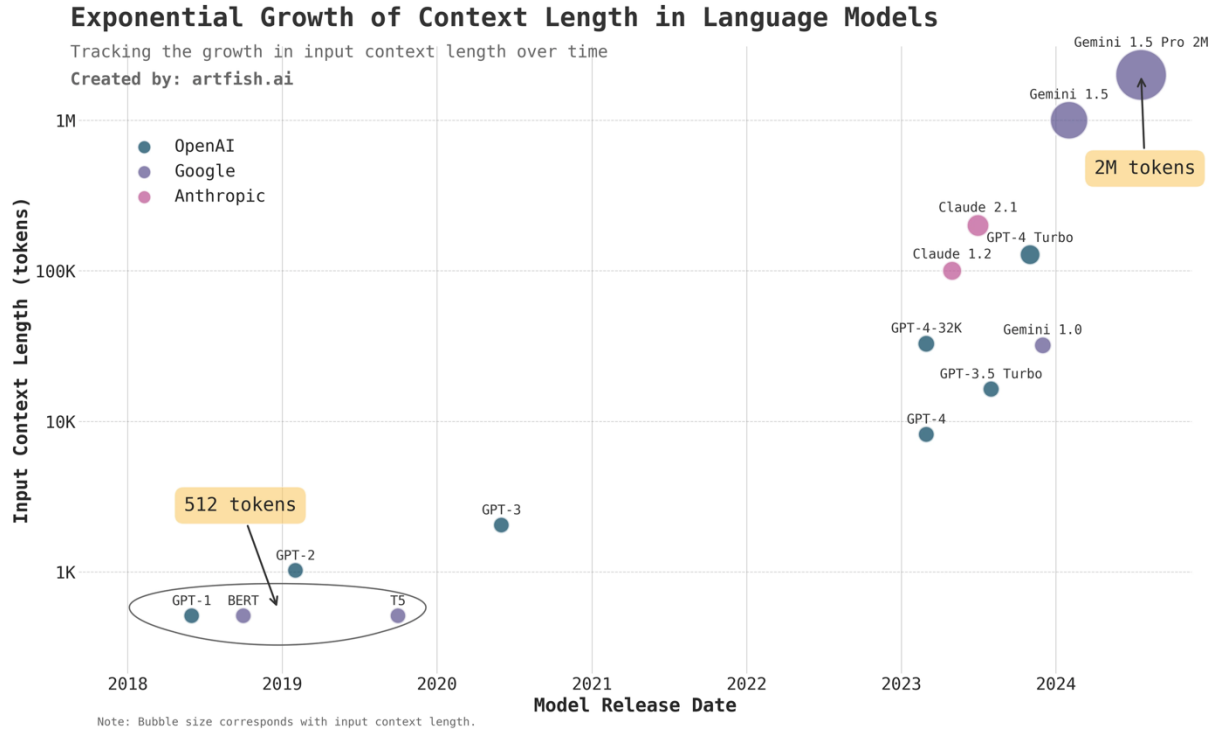
Context extension

„Easiest“ way is to add information directly to the prompt -> Extend the context



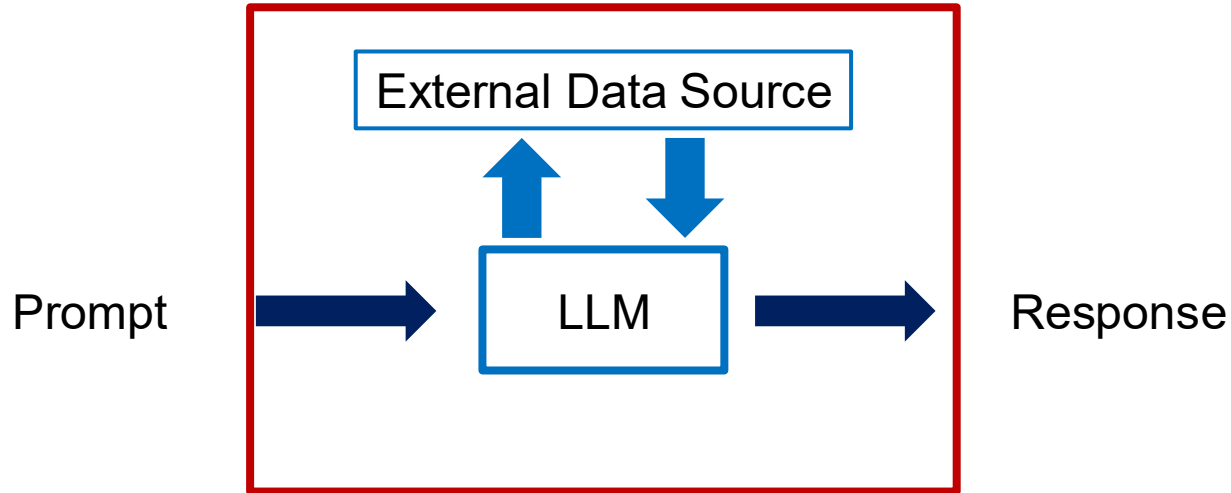
What could be problems with this approach?

Context extension



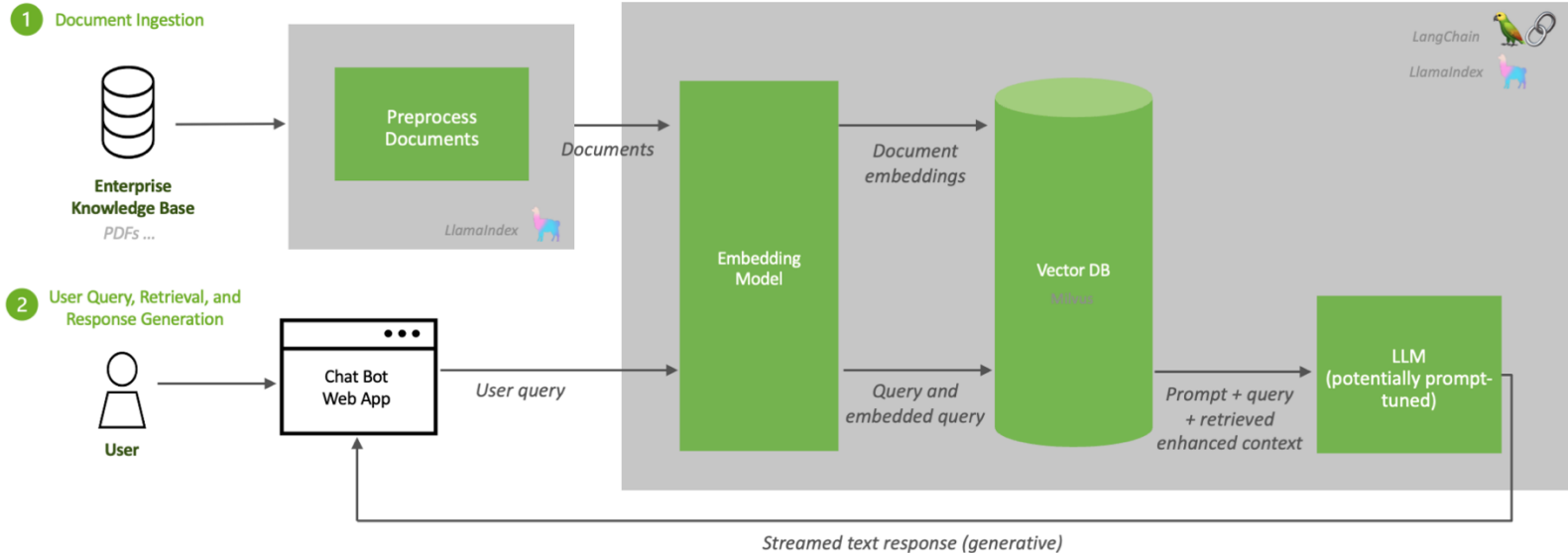
artfish.ai

Retrieval Augment Generation (RAG)



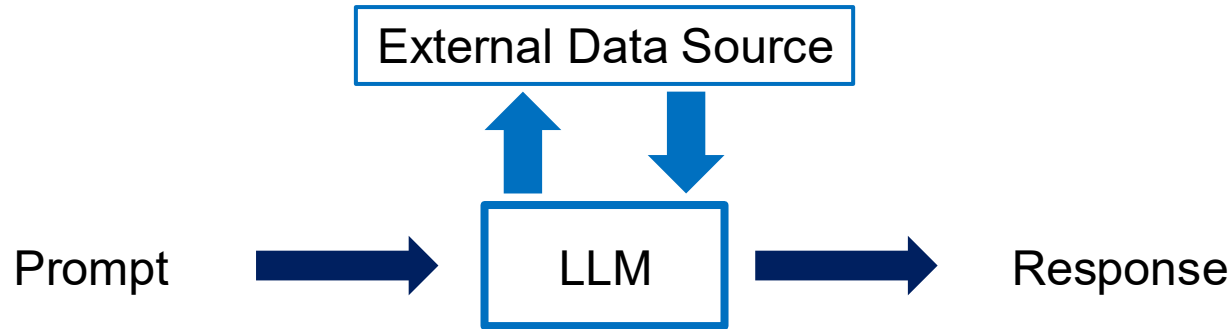
Retrieval Augment Generation (RAG)

Retrieval Augmented Generation (RAG) Sequence Diagram



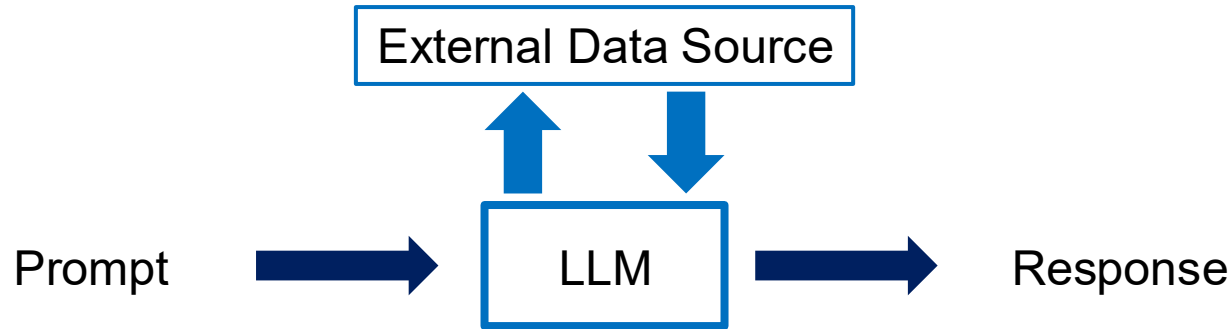
<https://developer.nvidia.com/blog/rag-101-demystifying-retrieval-augmented-generation-pipelines/>

Retrieval Augment Generation (RAG)



Demo + Exercise

Retrieval Augment Generation (RAG)



What could be problems with this approach?

LLMs with SAP

Single-tenant application (Retrieval Augmented Generation & Generative AI on SAP BTP)

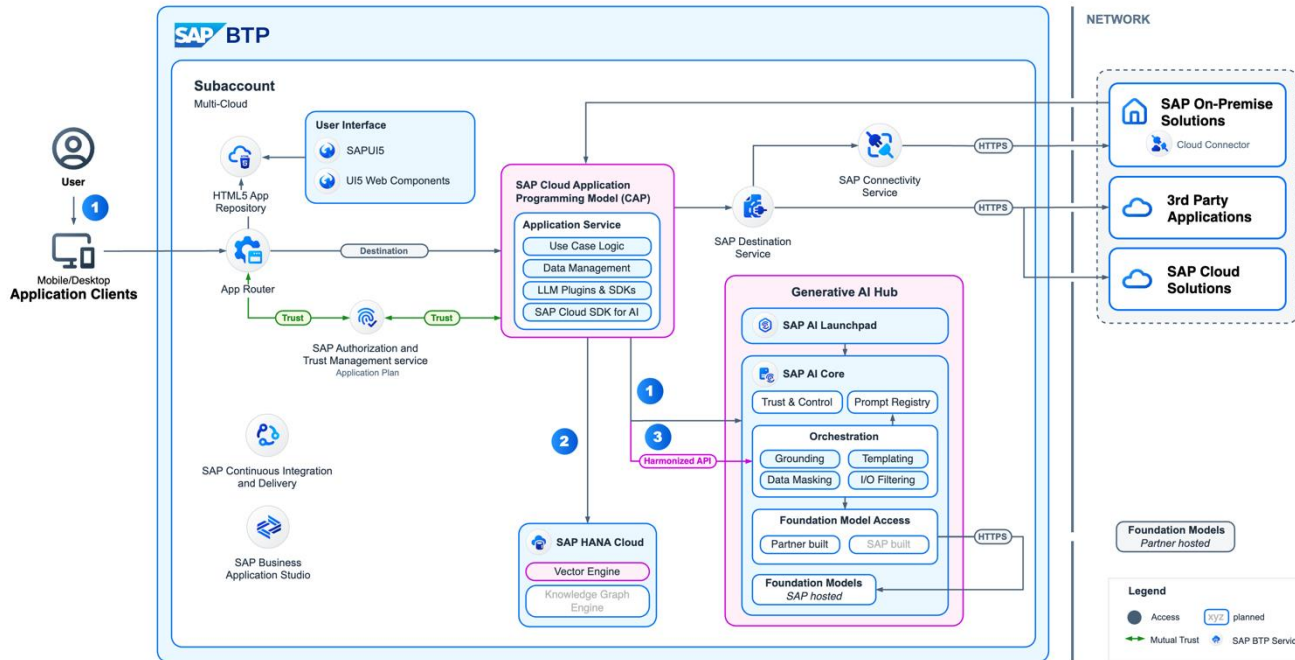


Diagram Level: L2

This BTP Solution Diagram depicts the reference architecture describing how to seamlessly combine various Large Language Models (LLMs) using the Generative AI Hub in SAP AI Core.

What is an AI agent?

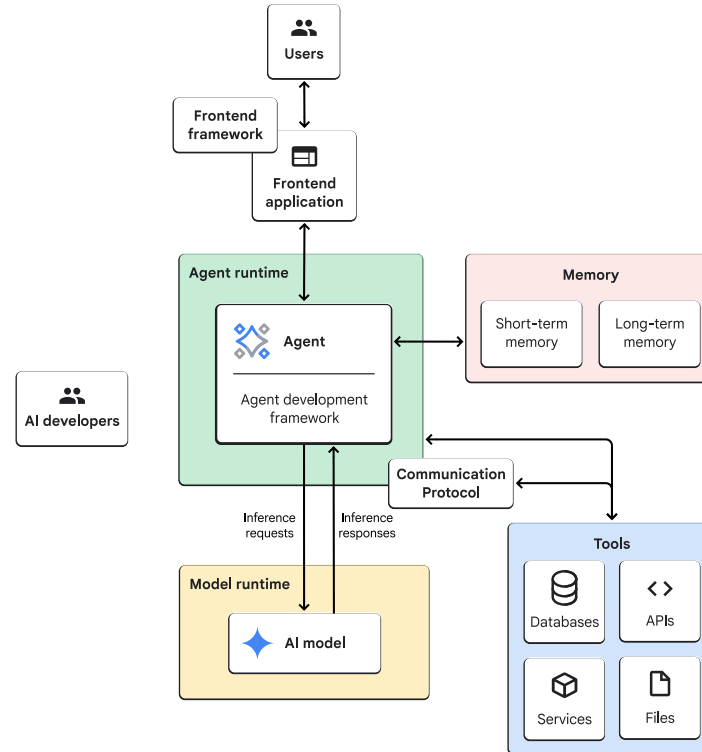
„Autonomous AI agents are technological entities that perceive, reason, and act upon information sensed from their environments while operating without external control in completing assigned tasks.

This definition clarifies two important advances highly relevant to sales scholarship and management:

- (1) it elevates the importance of reasoning in the AI's application cycle, and
 - (2) it distinguishes between systems that act with varying levels of autonomy.
- Of note, while often used interchangeably, the term “AI agent” more often refers to task-specific systems operating within predefined parameters, whereas “agentic AI” includes systems of multiple autonomous AI agents collaborating to achieve organizational objectives“

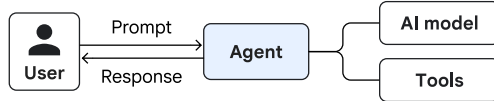
Gabriel R. Gonzalez, Johannes Habel, Gary K. Hunter,
AI agents, agentic AI, and the future of sales,
Journal of Business Research, Volume 202, 2026

AI Agent Components

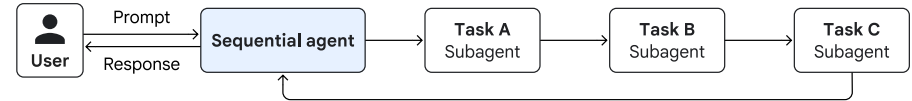


AI Agents

Single Agent System

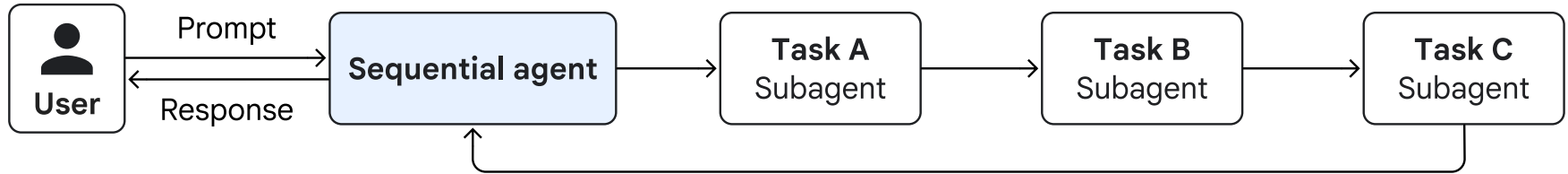


Multi Agent System



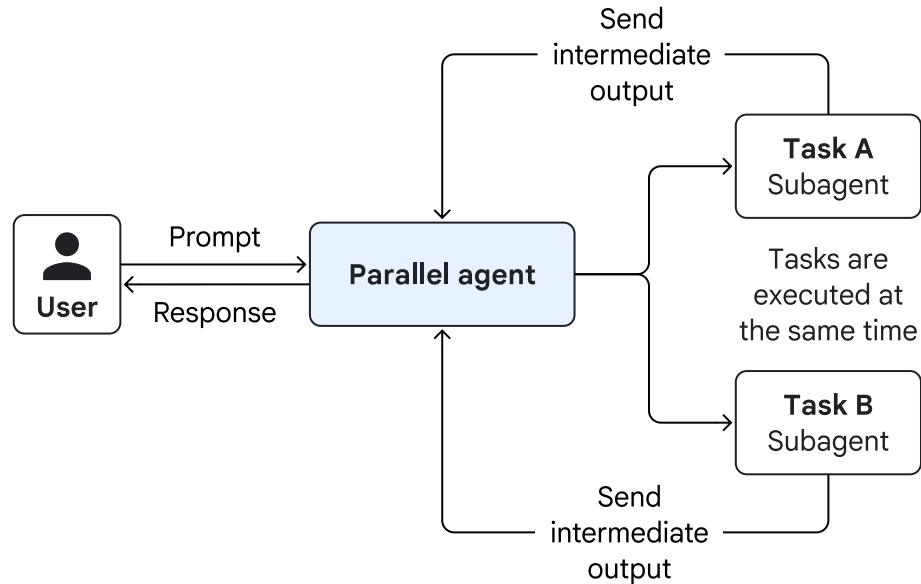
Multi Agent Patterns

Sequential pattern



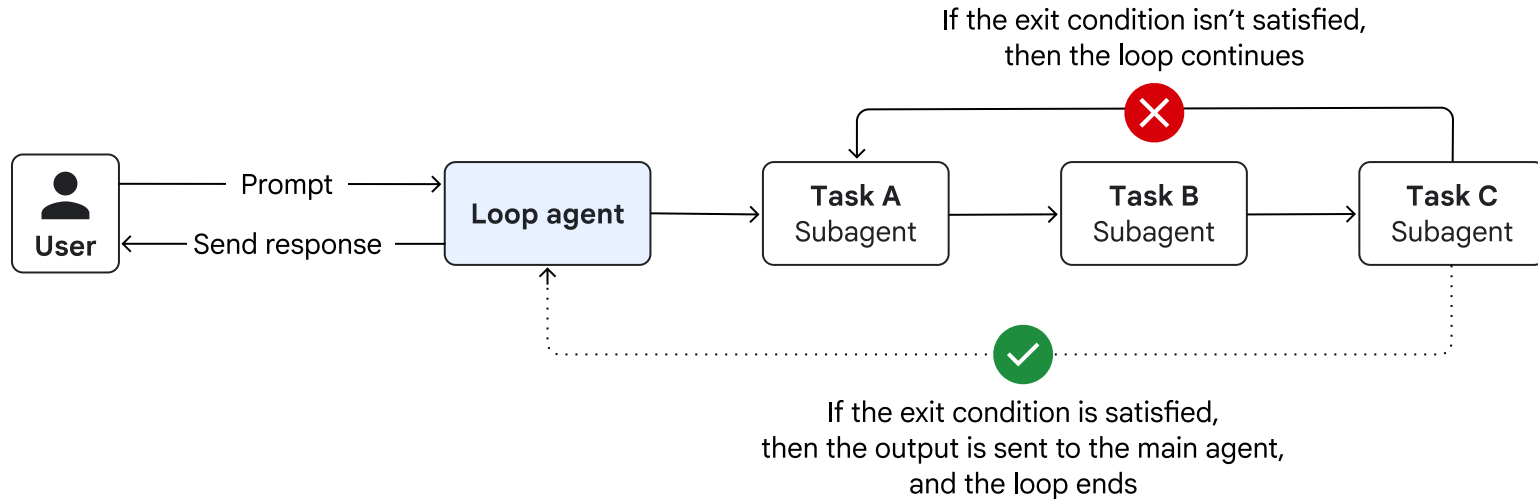
Multi Agent Patterns

Parallel pattern



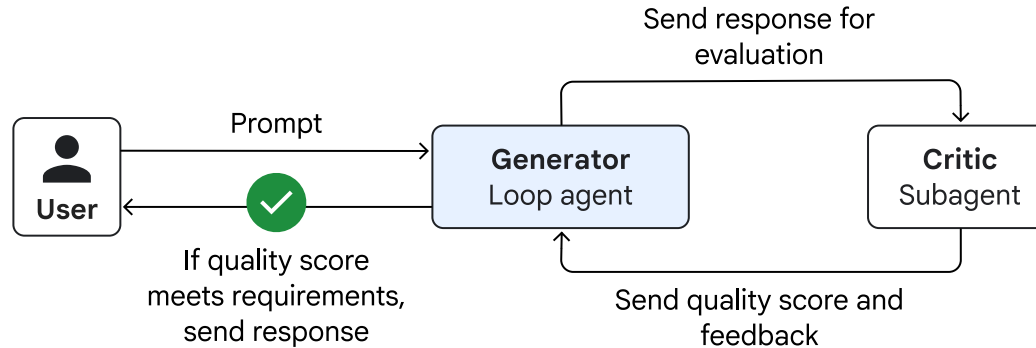
Multi Agent Patterns

Loop pattern



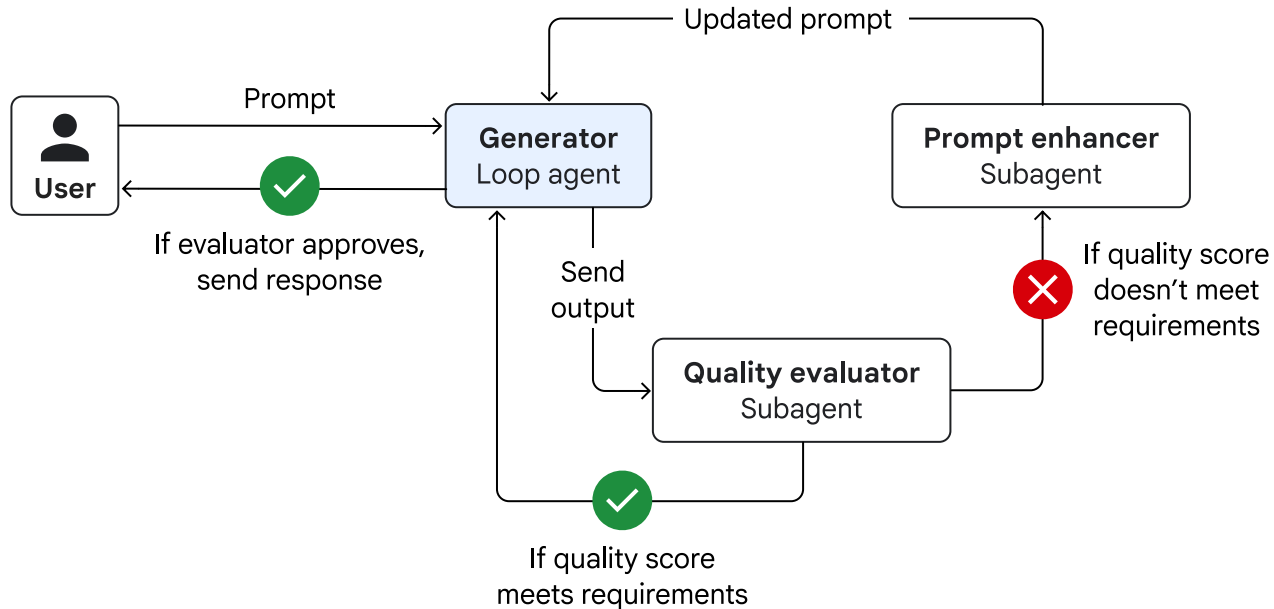
Multi Agent Patterns

Review and critique pattern



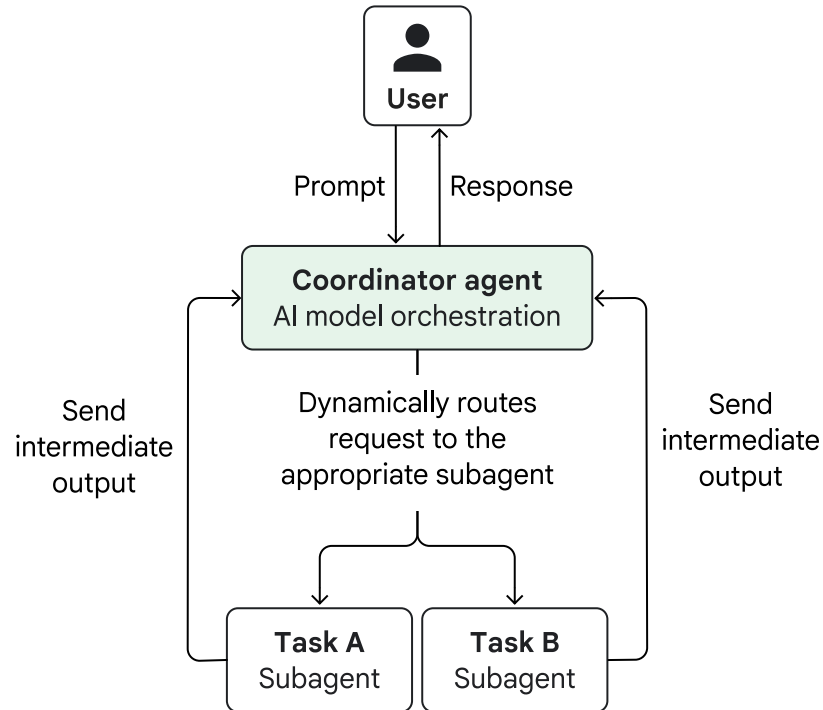
Multi Agent Patterns

Iterative refinement pattern



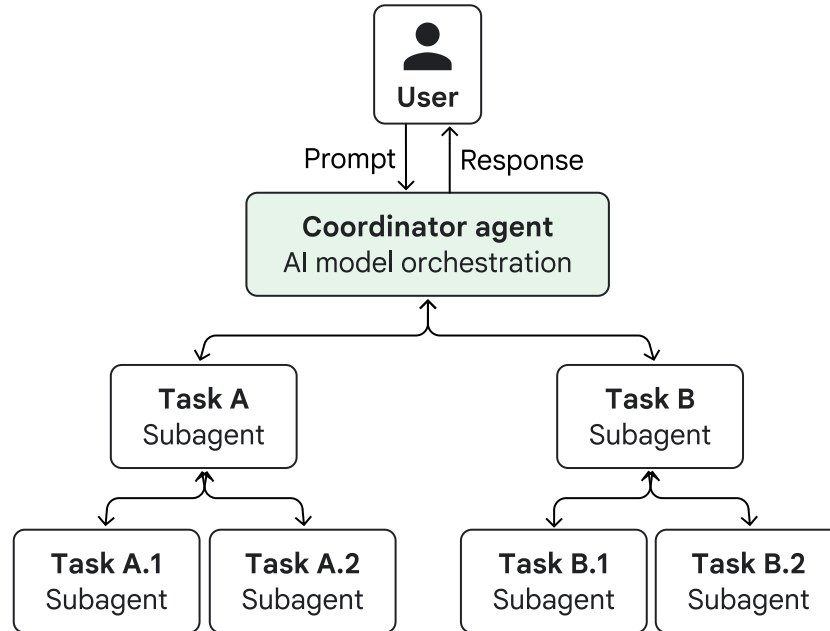
Multi Agent Patterns

Coordinator pattern



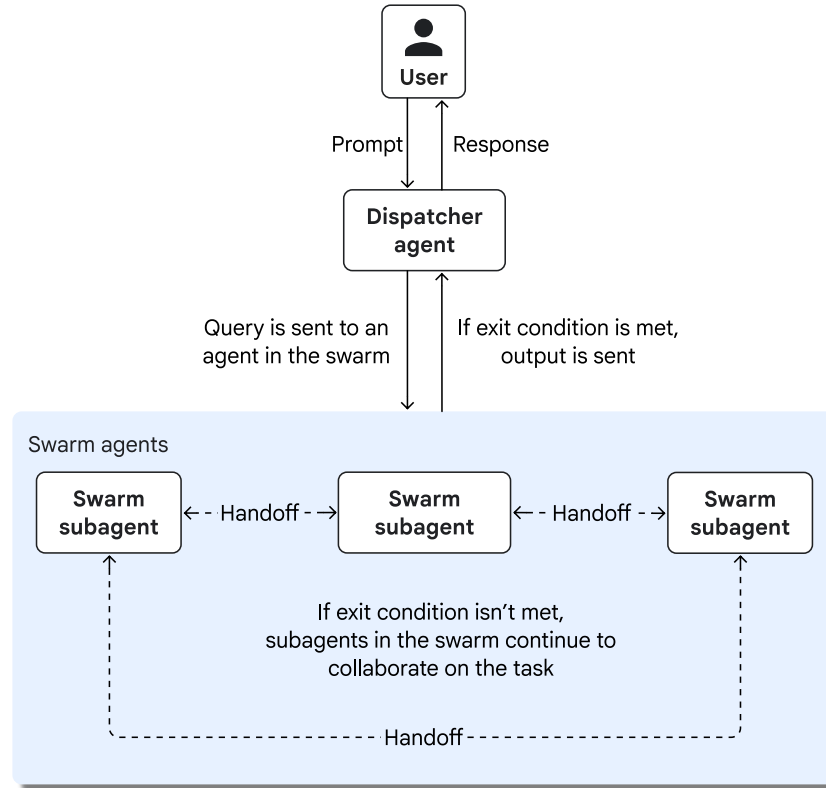
Multi Agent Patterns

Hierarchical task decomposition pattern (implementation of the coordinator pattern)



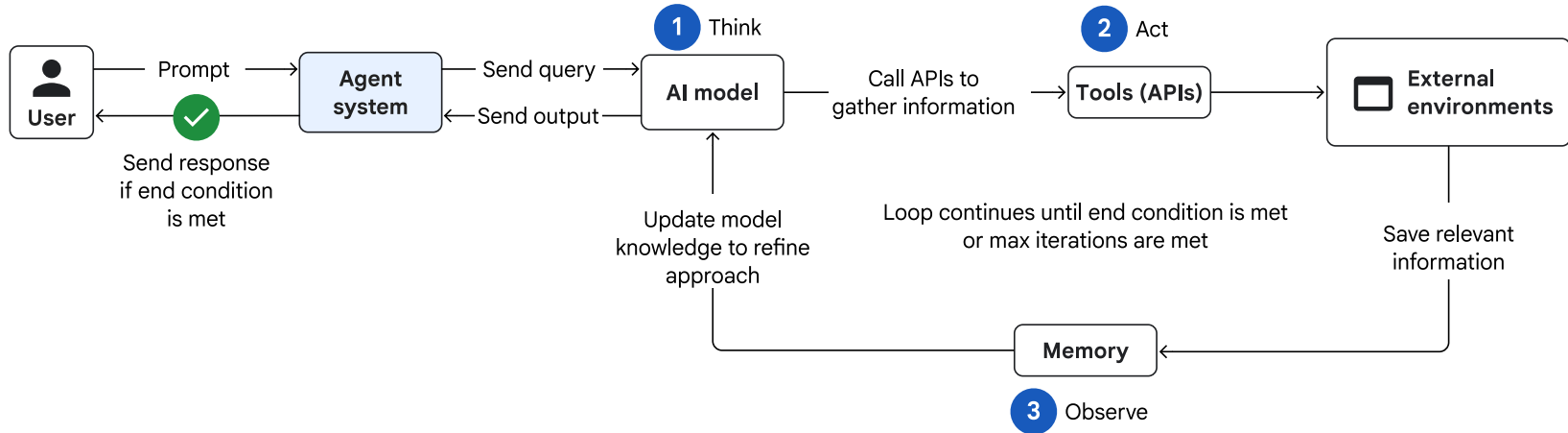
Multi Agent Patterns

Swarm pattern



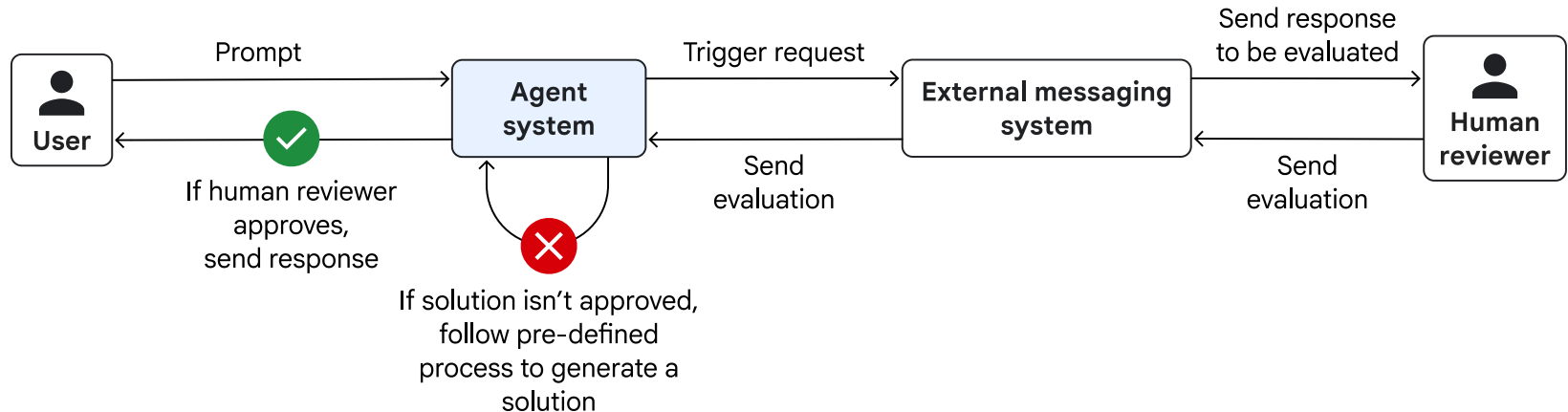
Agent Implementation Patterns

Reason and act (ReAct) pattern



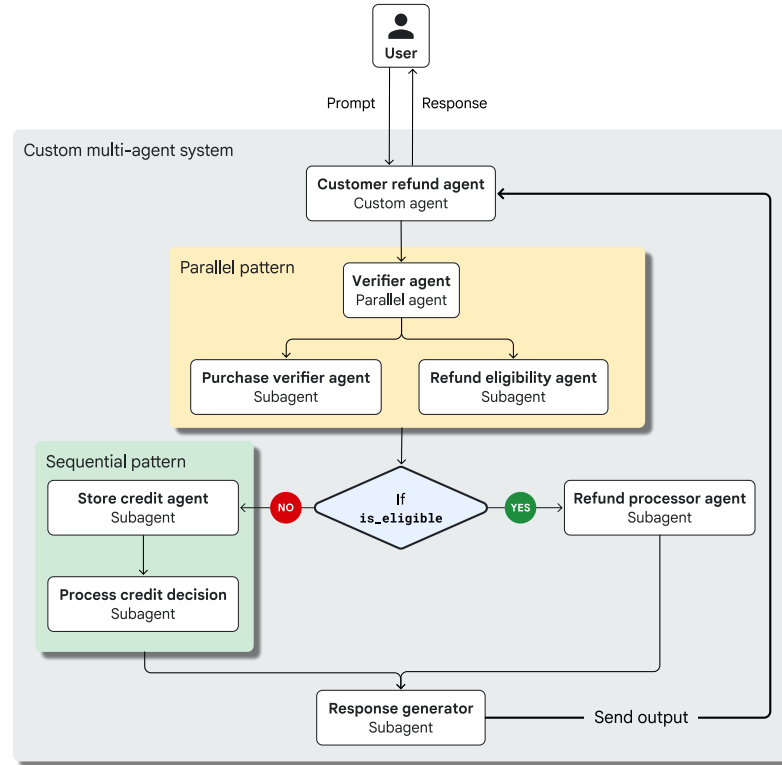
Agent Implementation Patterns

Human in the loop pattern



Agent Implementation Patterns

Custom logic pattern



Game Time



Develop your own agent

Literatur

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